

Notes: Benguria, Saffie, and Urzua (RES, 2024)

The Transmission of Commodity Price Super-Cycles

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1 Big picture

- **Question.** How do commodity price super-cycles transmit to the real economy of a commodity-producing emerging market?
- **Two transmission channels.** The paper separates a *wealth channel* from a *cost channel*. The wealth channel raises domestic demand when commodity prices rise. The cost channel raises wages when the commodity sector expands labour demand.
- **Empirical setting.** The paper studies Brazil during 1999–2013, a period with a large commodity boom followed by the start of a bust.
- **Identification strategy.** Brazil has strong regional variation in the composition of commodity production. The authors construct region-specific commodity price indexes using base-period commodity employment shares and world commodity price changes.
- **Cost channel evidence.** Commodity booms raise unskilled labour demand because the commodity sector is relatively unskilled-intensive. This lowers the skill premium and hurts relatively unskilled-intensive firms and industries.
- **Wealth channel evidence.** Commodity booms raise local income and spending, helping firms that sell locally: nonexporters and low-tradability manufacturing industries.
- **Granular data.** The empirical work uses administrative firm-level matched employer–employee data from Brazil, allowing the authors to study employment responses by sector, region, exporter status, and skill intensity.
- **Quantitative model.** The authors build a dynamic small open economy model with commodities, heterogeneous manufacturing firms, nontradables, skilled and unskilled workers, entry, export decisions, and an endogenous current account.
- **Main quantitative result.** A counterfactual economy in which the commodity boom is purely an endowment shock, with no labour-market transmission through the commodity sector, generates only about 30% of the intersectoral labour reallocation and about 40% of the within-tradable labour reallocation.
- **Welfare result.** The commodity super-cycle produces a consumption-equivalent welfare gain of 2.27% in the baseline economy, but 4.38% in the counterfactual economy where the labour-market cost channel is shut down.
- **Takeaway.** Commodity booms are not pure gifts. They raise wealth, but they also distort labour allocation through wage and relative-cost channels. These distortions reduce welfare gains from the boom.

2 Data and measurement

2.1 Brazil as the empirical setting

- Brazil is a large emerging market with substantial commodity exposure and substantial regional heterogeneity in commodity production.
- The sample period is 1999–2013, which includes a pronounced rise in world commodity prices and the beginning of a subsequent decline.
- The empirical design compares regions exposed to different commodity baskets, not simply Brazil before and after the boom.
- The analysis uses commuting zones as approximations to local labour markets.

Interpretation: The regional variation matters because the aggregate commodity boom is common to all Brazil. Identification comes from the fact that different regions were exposed to different commodities whose world prices moved differently.

2.2 Administrative firm-level data

- The main micro data come from RAIS, the Brazilian matched employer–employee administrative dataset.
- RAIS covers formal employment and reports information on firms, industries, workers, wages, education, and location.
- The authors classify workers into skilled and unskilled categories using educational attainment.
- The empirical analysis excludes government and quasi-public sectors, including federal and local government, education, health care, and domestic service.
- Firm sector and region are defined by where most of the firm’s employment is concentrated.

Interpretation: The dataset is rich enough to measure firm-level employment responses while also constructing region- and industry-level exposure to commodity price changes.

2.3 Economic sectors

The paper divides the economy into three broad sectors:

- **Commodity sector:** agriculture, mining, and fuels.
- **Tradable sector:** manufacturing industries.
- **Nontradable sector:** retail and wholesale trade, hospitality, construction, transportation, finance, real estate, and related services.

Interpretation: The commodity sector is the source of the shock. Tradables and nontradables are affected through costs, local demand, and export versus domestic sales exposure.

2.4 Skill intensity

Skill intensity is measured as the share of skilled workers in total industry employment:

$$SkillIntensity_i = \frac{L_i^S}{L_i^S + L_i^U}. \quad (1)$$

The paper documents three important patterns:

- commodity industries are relatively unskilled-intensive;
- tradable industries have higher skill intensity than commodities;
- nontradable industries display substantial heterogeneity in skill intensity.

Interpretation: Skill intensity is central to the cost channel. If commodity booms raise demand for unskilled labour, they lower the skill premium and change relative costs across industries.

2.5 Regional commodity price index

The regional commodity price index is a weighted average of individual commodity price changes:

$$Price_{r,t} = \sum_{i \in C} \lambda_{r,i,0} P_{i,t}, \quad (2)$$

where $\lambda_{r,i,0}$ is the base-period employment share of commodity i in region r , and $P_{i,t}$ is the price of commodity i .

The empirical regressions often use a residualized version of this index after removing region and time fixed effects.

Interpretation: The index is a sufficient statistic for regional exposure to the commodity boom in the conceptual framework. It lets the authors compare regions exposed to different commodity price paths.

2.6 Exporter status and tradability

- Exporter status is measured at the firm level inside manufacturing.
- A firm-level exporter dummy equals one if the firm exports in the relevant period.
- Tradability is an industry-level measure based on the extent to which output is sold outside local markets.
- Low-tradability industries are more locally oriented; high-tradability industries are more exposed to national or foreign markets.

Interpretation: Exporter status and tradability are the empirical tools for identifying the wealth channel. If local demand rises, local sellers should benefit more than exporters.

3 Conceptual framework and testable implications

3.1 Sectoral structure

The analytical framework is a specific-factors model with regions r and industries i . Skilled and unskilled labour are mobile across industries within a region, while the specific factor is immobile across industries and regions. The economy has a commodity sector C , two nontradable industries, and one tradable industry.

The key skill-intensity ordering is:

$$\frac{\theta_s^C}{\theta_u^C} < \frac{\theta_s^T}{\theta_u^T} = \frac{\theta_s^{N\ell}}{\theta_u^{N\ell}} < \frac{\theta_s^{Nh}}{\theta_u^{Nh}}. \quad (3)$$

Interpretation: Commodities are the most unskilled-intensive, tradables and low-skill nontradables have the same skill intensity, and high-skill nontradables are the most skilled-intensive.

3.2 Proposition 1: regional commodity prices and the skill premium

The paper shows that the change in the regional skill premium can be approximated by:

$$\hat{w}^S - \hat{w}^U \approx \kappa_C \left(\sum_{i \in C} \lambda_i \hat{P}_i \right), \quad \kappa_C < 0. \quad (4)$$

Interpretation: Because the commodity sector is unskilled-intensive, a commodity price increase raises demand for unskilled labour and lowers the skill premium.

3.3 Testable implication 1: skill premium

$$\sum_{i \in C} \lambda_{r,i} \hat{P}_{i,t} \uparrow \Rightarrow \hat{w}_{r,t}^S - \hat{w}_{r,t}^U \downarrow. \quad (5)$$

Interpretation: Regions facing stronger commodity price increases should experience larger declines in the skill premium.

3.4 Proposition 2: cost channel

The relative employment change between high- and low-skill nontradable industries is:

$$\hat{L}^{Nh} - \hat{L}^{N\ell} = (\eta_U^{Nh} - \eta_U^{N\ell})(\hat{w}^S - \hat{w}^U). \quad (6)$$

Using Proposition 1:

$$\hat{L}^{Nh} - \hat{L}^{N\ell} \approx (\eta_U^{Nh} - \eta_U^{N\ell})\kappa_C \left(\sum_{i \in C} \lambda_i \hat{P}_i \right). \quad (7)$$

Interpretation: If unskilled wages rise relative to skilled wages, low-skill industries face a larger cost increase. Therefore high-skill industries grow relatively more.

3.5 Testable implication 2: firm-level employment by skill intensity

$$Price_{r,t} \times SkillIntensity_i \text{ should have a positive coefficient in employment regressions.} \quad (8)$$

Interpretation: In regions with larger commodity booms, high-skill firms or industries should outperform low-skill firms or industries.

3.6 Proposition 3: wealth channel

For the tradable industry and a low-skill nontradable industry with the same technology, the model implies:

$$\hat{L}^T - \hat{L}^{N\ell} = -\hat{m}, \quad (9)$$

where $\hat{m}$ is the change in regional income. Since regional income rises with commodity prices:

$$\hat{m} = \sum_i \phi_i \hat{P}_i. \quad (10)$$

Interpretation: Higher local income raises demand for locally sold goods. Domestic sellers benefit more than exporters.

3.7 Testable implication 3: exporters versus local sellers

$$Price_{r,t} \times Exporter_{f,t} \text{ should have a negative coefficient in manufacturing employment regressions.} \quad (11)$$

Interpretation: Exporters sell to outside markets and benefit less from local demand booms. Nonexporters and low-tradability industries benefit more from the wealth channel.

4 Empirical specifications

4.1 Skill premium regression

The empirical test of Proposition 1 is:

$$SP_{r,t} = \beta_1 Price_{r,t} + \gamma_r + \delta_t + \varepsilon_{r,t}, \quad (12)$$

where $SP_{r,t}$ is the regional skill premium. Some specifications use annual earnings and others use monthly December earnings.

Interpretation: The coefficient β_1 should be negative. Table 1 confirms this prediction.

4.2 Cost-channel firm-level regression

The main cost-channel regression is:

$$\log(Emp)_{f,t} = \alpha_0 Price_{r,t} + \alpha_c [Price_{r,t} \times SkillIntensity_i] + \gamma_f + \delta_{s,t} + \varepsilon_{f,t}. \quad (13)$$

- f indexes firms.
- r is the region of the firm.
- i is the firm's industry.
- γ_f are firm fixed effects.
- $\delta_{s,t}$ are state-by-time fixed effects.
- Standard errors are clustered by region.

Interpretation: The coefficient α_c is the cost-channel parameter. It should be positive because high-skill industries are hurt less by unskilled wage increases.

4.3 Wealth-channel firm-level regression

The main wealth-channel regression is:

$$\log(Emp)_{f,t} = \beta_0 Price_{r,t} + \beta_1 Exporter_{f,t} + \beta_{w1} [Exporter_{f,t} \times Price_{r,t}] \quad (14)$$

$$+ \beta_{w2} [Tradability_i \times Price_{r,t}] + \gamma_f + \delta_{s,t} + \mu_{f,t}. \quad (15)$$

Interpretation: The coefficients $\beta_{w1} < 0$ and $\beta_{w2} < 0$ indicate that exporters and highly tradable industries benefit less from local wealth increases.

4.4 Identification threats and robustness

The paper discusses and checks many alternative mechanisms:

- upstream and downstream linkages;
- interest-rate and financial-condition heterogeneity;
- import competition and China shock exposure;
- exchange rate effects;
- local minimum wages;
- HP-filtered commodity price cycles;

- multiway correlated standard errors;
- commodity supply shocks and country-level endogeneity of world prices;
- spatial spillovers across regions.

Interpretation: The main cost and wealth channel estimates remain robust across these alternative explanations.

5 Quantitative model

5.1 Representative household

The household supplies skilled and unskilled labour inelastically and consumes a bundle of imported, commodity, tradable, and nontradable goods. Lifetime utility is:

$$U = \sum_{t=0}^{\infty} \beta^t \frac{C_t^{1-\nu}}{1-\nu}. \quad (16)$$

The consumption bundle is a nested CES aggregate of imports M_t , tradables T_t , nontradables N_t , and domestic commodities C_t^d .

Interpretation: The household smooths consumption intertemporally, so the commodity boom works partly through permanent-income wealth effects and current account dynamics.

5.2 Budget constraint and stochastic discount factor

The household can save internationally through bonds. The stochastic discount factor satisfies:

$$\Psi_{t,t+1} = \beta \frac{\lambda_{t+1}}{\lambda_t} = \frac{1}{1+r}. \quad (17)$$

Interpretation: After a surprise change in commodity prices, the household adjusts consumption immediately to a new present value of income.

5.3 Labour composite

Firms combine skilled and unskilled labour through a CES aggregator:

$$\ell_{f,t}^k = \left[(\phi^k)^{1/\gamma} (\ell_{f,t}^{u,k})^{(\gamma-1)/\gamma} + (1-\phi^k)^{1/\gamma} (\ell_{f,t}^{s,k})^{(\gamma-1)/\gamma} \right]^{\gamma/(\gamma-1)}. \quad (18)$$

The effective labour cost in sector k is:

$$\hat{w}_t^k = \left[\frac{\phi^k}{(w_t^u)^{\gamma-1}} + \frac{1-\phi^k}{(w_t^s)^{\gamma-1}} \right]^{-1/(\gamma-1)}. \quad (19)$$

Interpretation: A fall in the skill premium lowers costs for skilled-intensive sectors relative to unskilled-intensive sectors.

5.4 Commodity sector

The commodity sector produces:

$$Y_t^C = (L_t^C)^\eta (R_t)^{1-\eta}, \quad (20)$$

where R_t is a resource input. The commodity price P_t^C is exogenous in world markets.

Interpretation: The commodity sector is the origin of the external price shock and the labour demand shock.

5.5 Nontradable firms

A nontradable firm with productivity z produces:

$$q_{Nt}(z) = z(\ell_{Nt}(z) - f_N). \quad (21)$$

Firms draw productivity upon entry and face exit risk. Entry has fixed and convex components:

$$f_{Net} = f_{Ne} + \xi_{Ne} \left[\exp \left(\frac{M_{Net} - M_{Ne}}{M_{Ne}} \right) - 1 \right]. \quad (22)$$

Interpretation: The nontradable sector absorbs local demand booms and is central to the wealth channel.

5.6 Tradable firms and export cutoff

Tradable firms produce:

$$q_{Tt}(z) = z(\ell_{Tt}(z) - f_d). \quad (23)$$

Exporting requires a per-period fixed cost f_x and faces foreign demand:

$$q_{Tt}^x(z) = \Gamma(p_{Tt}(z))^{-\sigma}. \quad (24)$$

Interpretation: Heterogeneous productivity creates an endogenous exporter cutoff, allowing the model to match the empirical exporter versus nonexporter wealth-channel evidence.

5.7 Entry, exit, and equilibrium

The value function for a nontradable firm is:

$$W_t(z) = \max \{0, \pi_t(z) + \beta(1 - \delta_N)W_{t+1}(z)\}. \quad (25)$$

The free-entry condition is:

$$\widehat{w}_{Nt} f_{Net} = \int_{\bar{z}_t}^{\infty} W_t(z) g(z) dz. \quad (26)$$

For tradable firms, domestic and export values are:

$$W_{d,t}(z) = \max \{0, \pi_{d,t}(z) + \beta(1 - \delta_T)W_{t+1}(z)\}, \quad (27)$$

$$W_{x,t}(z) = \max \{0, \pi_{d,t}(z) + \pi_{x,t}(z) + \beta(1 - \delta_T)W_{t+1}(z)\}. \quad (28)$$

Interpretation: This structure links commodity-price shocks to entry, exit, exporting, and labour reallocation.

6 Calibration, validation, and counterfactuals

6.1 Calibration

- The model is calibrated to Brazil in 2001, just before the boom.
- Externally calibrated parameters come from macroeconomic data, RAIS firm-level data, and the literature.
- Internally calibrated parameters target sector shares, export shares, firm-size moments, skill premium, skill allocations, and entry-cost moments.
- The commodity price path used in the model mimics the empirical bust-to-boom path from 2002 to 2013.

Interpretation: The calibration forces the model to match the firm, sector, trade, and labour-market features that matter for the transmission channels.

6.2 Model validation

The authors validate the model by estimating empirical-style regressions in simulated data and comparing them to the actual data.

- Table 6 compares the model and data for the skill premium and tradable versus nontradable employment.
- Table 7 compares the model and data for exporter versus nonexporter employment in the tradable sector.
- The model closely matches the signs and broad magnitudes of the empirical transmission-channel coefficients.

Interpretation: The model is not just calibrated to macro moments. It also reproduces the micro regression evidence that motivated the two channels.

6.3 Counterfactual economy

The main counterfactual shuts down the commodity sector’s labour-demand response to commodity price fluctuations, making the commodity boom a pure endowment shock.

Interpretation: Comparing the baseline and counterfactual isolates the role of labour-market transmission in reducing welfare and causing reallocation.

7 Tables and figures

This section follows the *Read* plus *Economic message* structure. All visuals are directly cropped from the paper.

7.1 Figure 1: skill intensity across industries

Read:

- Commodity industries have the lowest share of skilled employment.
- Tradable industries are more skilled-intensive than commodities.
- Nontradable industries have a wide distribution of skill intensity.

Economic message: This ordering is the empirical basis for the cost channel: a commodity boom raises unskilled labour demand and shifts relative costs across industries.

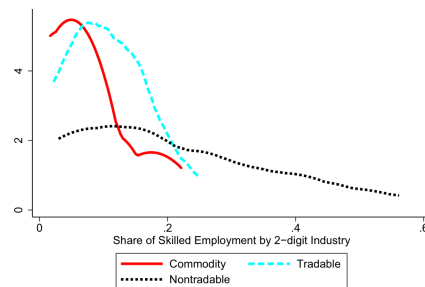


FIGURE 1
Skill intensity across industries
s figure displays kernel density estimates of the distribution of skill intensity (measured as the share of skilled w
wo-digit industries in the commodity, tradable, and nontradable sectors.

7.2 Figure 2 and Tables 1–3: empirical transmission channels

Read:

- Figure 2 documents the commodity super-cycle and regional variation in commodity prices.
- Table 1 shows that higher regional commodity prices lower the skill premium; coefficients are about -0.140 and -0.128 .
- Table 2 shows the cost channel. The interaction between skill intensity and commodity prices is positive and significant in all sectors.
- Table 3 shows the wealth channel. Exporters and high-tradability industries have lower employment growth during regional commodity booms.

Economic message: The empirical evidence supports both channels: commodity booms change relative wages and raise local demand.

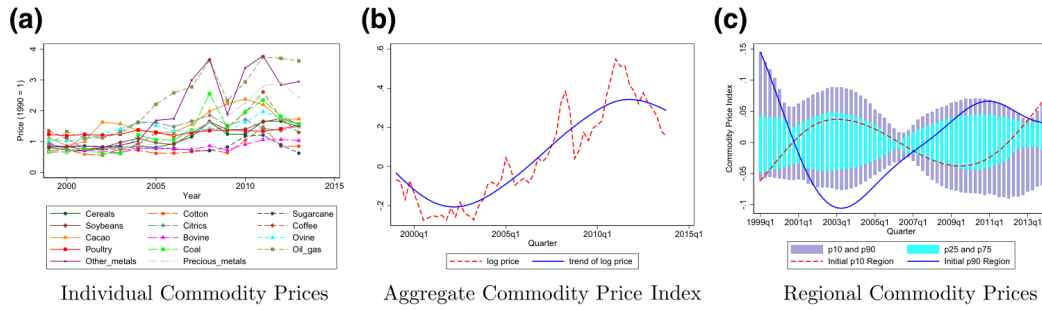


FIGURE 2
Commodity prices

Note: The graph on the left shows the path of the prices in real U.S. dollars of the fourteen commodities included in our commodity price index. These prices are normalized to one in 1990. The graph on the middle shows the path of the commodity price index for Brazil (dashed line) and its trend (solid line). The graph on the right shows the percentiles of the distribution of the residual regional commodity price index after extracting region and time-fixed effects. The outer bars mark the tenth and ninetieth percentiles and the inner bars mark the twenty-fifth and seventy-fifth percentiles. The dashed (solid) line marks the path of the residual price for the region at the tenth (ninetieth) percentile in the initial period.

	(1)	(2)
$Price_{it}$	-0.140^{**} (0.057)	-0.128^{***} (0.058)
Observations	8363	7912
R^2	0.694	0.687

Notes: This table reports the results of the estimation of Equation (1). Column 1 corresponds to annual earnings. Column 2 corresponds to monthly (December) earnings. Each regression includes region as well as macroregion $\times$ time (year) fixed effects. Standard errors are clustered by region. ***, **, and * denote statistical significance at a 1, 5, and 10% confidence level, respectively.

Table 1: skill premium

	Nontradable (1)	Tradable Sector (2)	Commodity Sector (3)
$Price_{it}$	0.014 (0.021)	-0.136^{***} (0.030)	0.190* (0.096)
Skill Intensity $\times Price_{it}$	0.845^{***} (0.114)	0.795^{***} (0.205)	3.467^{***} (0.476)
Observations	30,453,457	8,007,422	1,007,570
R^2	0.849	0.873	0.908

Notes: This table reports the results of the estimation of Equation (4) by sector. Each regression includes firm and state $\times$ time (year-quarter) fixed effects. Note we estimate Equation (2) demeaning the commodity price index and skill

Table 2: cost channel

	(1)	(2)	(3)
$Price_{it}$	-0.042^* (0.022)	-0.046^{**} (0.022)	-0.044^{**} (0.022)
Exporter f_{it}	0.135^{***} (0.005)	0.135^{***} (0.005)	0.134^{***} (0.005)
Exporter $f_{it} \times Price_{it}$	-0.008^{***} (0.015)	-0.008^{***} (0.015)	-0.005^{***} (0.015)
Tradability $\times Price_{it}$	-0.042^{**} (0.017)	-0.042^{**} (0.017)	-0.039^{**} (0.017)
Observations	8,086,964	8,005,174	8,005,174
R^2	0.909	0.909	0.909

Table 3: wealth channel

7.3 Tables 4 and 5: calibrated parameters

Read:

- Table 4 reports externally calibrated parameters, including interest rate, discount factor, substitution elasticities, labour shares, and exit rates.
- Table 5 reports internally calibrated parameters, including sector weights, export fixed costs, productivity distributions, foreign demand, resource endowment, skill shares, and entry-cost elasticity.
- The targeted moments include sector shares, export share, firm-size shares, skill premium, and skill allocation across sectors.

Economic message: The calibration embeds the sectoral, firm-level, and labour-market heterogeneity needed to quantify the two transmission channels.

Externally calibrated parameters			
Parameter	Description	Value	Source
r^*	Interest rate	0.04	Macroeconomic data
ν	Discount factor	1.5	Literature
σ	Elast. of substitution between varieties in T and N	4	Simonovska and Waugh (2014)
ρ	Inverse of markup in T and N	0.75	$\sigma = \frac{1}{1-\rho}$
γ	Elast. of subs. between skilled and unskilled labour	1.41	Katz and Murphy (1992)
η	Labour share in C	0.5	Gollin <i>et al.</i> (2014)
δ_T	Exogenous exit probability in T	0.14	Firm-level data (RAIS)
δ_N	Exogenous exit probability in N	0.18	Firm-level data (RAIS)
θ_1	Elast. of subs. between importables and domestic tradables	$\frac{1}{1-0.85}$	Corsetti <i>et al.</i> (2008)
θ_2	Elast. of subs. between tradables and nontradables	$\frac{1}{1-0.74}$	Mendoza (1995)
θ_3	Elast. of subs. between commodities and non-commodities	$\frac{1}{1-0.50}$	Buara and Kaboski (2009)
μ_T	Mean of productivity distribution in T	0	Normalization
$\bar{B}$	Steady-state debt	0	Normalization
$\bar{H}$	Total inelastic labour supply	100	Normalization
a_T	Weight on T good in utility function	1	Normalization
p_T	Steady-state price of T good	1	Normalization

Table 4: externally calibrated parameters

Internally calibrated parameters					
Parameter	Description	Moment Targeted	Moment (Data)	Moment (Model)	Param. Value
P_C	Commodity price	C share in GDP	0.14	0.13	26.86
a_N	Weight on N good in utility	N share in GDP	0.55	0.56	8.40
a_C	Weight on C good in utility	Share of C in exports	0.32	0.34	0.54
f_d	Production fixed cost in T	Share of T emp. in 20% smallest firms	0.02	0.03	0.06
f_x	Fixed cost of exporting in T	Fraction of exporting firms in T	0.09	0.08	1.40
Σ_T	St. dev. of productivity distribution in T	Share of T emp. in 10% largest firms	0.68	0.71	0.91
Γ	Foreign demand	Export to GDP ratio	0.15	0.19	281.16
$\bar{R}$	Resource endowment	C share in employment	0.08	0.08	3.63
f_N	Production fixed cost in N	Share of N emp. in 20% smallest firms	0.04	0.03	0.03
μ_N	Mean of productivity distribution in N	Relative mass of firms: M_T/M_N	0.27	0.27	0.25
Σ_N	St. dev. of productivity distribution in N	Share of N emp. in 10% largest firms	0.66	0.56	0.79
κ	Aggregate share of unskilled labour	Skill premium (w^S/w^U)	3.16	3.16	0.83
ϕ^C	Skill intensity in C	Shared of unskilled labour in C ($L^U \cdot C / L^C$)	0.92	0.92	0.69
ϕ^N	Skill intensity in N	Shared of unskilled labour in N ($L^U \cdot N / L^N$)	0.79	0.79	0.43
ϕ^T	Skill intensity in T	Shared of unskilled labour in T ($L^U \cdot T / L^T$)	0.88	0.88	0.59
ζ_e	Entry congestion elasticity	Fraction of entry cost: $\max \left\{ \frac{\text{cong. term}}{f_e} \right\}$	0.20	0.19	1

Notes: This table shows the values of internally calibrated parameters. The abbreviations C, T, and N stand for the commodity, tradable, and nontradable sectors, respectively.

Table 5: internally calibrated parameters

7.4 Figure 3 and Tables 6–7: model validation

Read:

- Figure 3 shows the commodity price super-cycle used in the quantitative model.
- Table 6 validates the model against the empirical skill premium and regional employment regressions.
- Table 7 validates the model against the empirical exporter-versus-nonexporter employment response.

Economic message: The simulated economy reproduces the main empirical elasticities, supporting the model's use for counterfactual and welfare analysis.

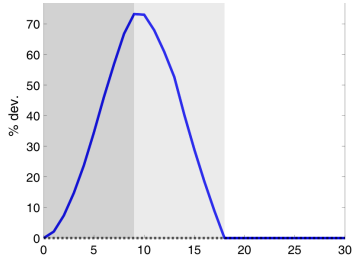


FIGURE 3
Commodity price super-cycle

shows the path of the price of the commodity good used in the quantitative analysis.

TABLE 6
Figure 3: commodity price super-cycle

Model validation: skill premium and regional employment				
	(1)	(2)	(3)	(4)
	Skill premium		Regional employment	
	Model	Data	Model	Data
Price	−0.085*** (0.003)	−0.221*** (0.087)		
Price $\cdot T_S$			−0.053*** (0.002)	−0.199*** (0.026)
Observations	11	267	22	594

Notes: The first two columns report the results of the estimation of Equation (12). Column 1 includes year-fixed effects; Column 2 includes state- and year-fixed effects. The last two columns report the results of the estimation of Equation (13). Column 3 includes sector and year-fixed effects; Column 4 includes sector and state $\times$ year-fixed effects. Columns 1 and 3 use simulated data. Columns 2 and 4 use the actual data. ***, **, and * denote statistical significance at a 1, 5, and 10% confidence level, respectively.

Table 6: model validation, skill premium and regional employment

Model validation: firm-level employment		
	(1)	(2)
	Tradable Sector	
	Model	Data
Exporter $f_T \times \text{Price}_{eT}$	−0.264*** (0.000)	−0.094*** (0.024)
Observations	1,315,872	1,538,219

Notes: Columns 1 and 2 in this table report the results of the estimation of Equation (14) using the simulated and actual data, respectively. ***, **, and * denote statistical significance at a 1, 5, and 10% confidence level, respectively.

Table 7: model validation, firm-level employment

7.5 Figures 4 and 5: wage dynamics and labour reallocation

Read:

- Figure 4 shows that the baseline boom raises unskilled wages and lowers the skill premium sharply.
- In the counterfactual economy, wage dynamics are muted because the commodity sector's labour-demand shock is shut down.
- Figure 5 shows that the baseline generates much stronger intersectoral reallocation between tradables and nontradables.
- Figure 5 also shows stronger within-tradable reallocation toward domestic production relative to exports.

Economic message: The labour-market cost channel accounts for most of the reallocation caused by the commodity boom.

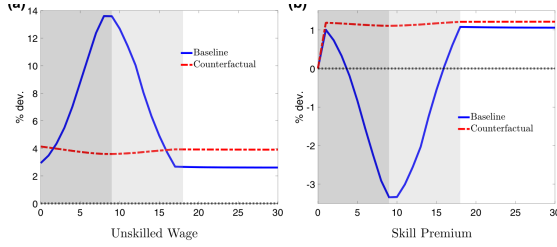


FIGURE 4

Wage dynamics in the baseline and counterfactual economies

Note: This figure shows the response of the unskilled wages and the skill premium to the commodity price super-cycle shown in Figure 3. The blue solid line corresponds to the baseline economy and the red dashed line corresponds to the counterfactual economy described in Section 4.3.

construction, the commodity super-cycle has no direct impact on wages. Second-order wage

Figure 4: wage dynamics

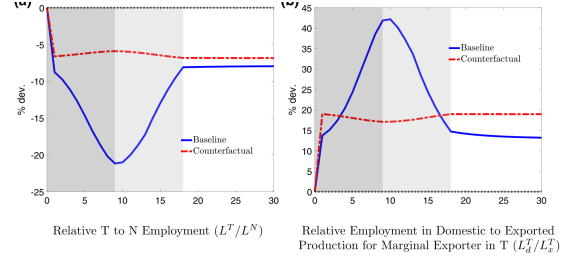


FIGURE 5

Employment reallocation in the baseline and counterfactual economies

Note: This figure shows the response of the relative tradable to nontradable sector employment (left) and the ratio of labour employed in the production of tradables for the domestic market to labour employed in the production of tradables for the foreign market in the marginal exporter defined in the text (right) to the commodity price super-cycle shown in Figure 3. The blue solid line corresponds to the baseline economy and the red dashed line corresponds to the counterfactual economy described in Section 4.3.

resulting from the anticipated boom, while only the baseline economy faces a gradual reallo-

Figure 5: employment reallocation

7.6 Figure 6 and Table 8: macro responses and welfare

Read:

- Figure 6 shows that the counterfactual economy has a much larger GDP boom and larger wealth accumulation.
- The baseline economy experiences stronger real exchange rate appreciation and smaller welfare gains.
- Table 8 reports a baseline consumption-equivalent welfare gain of 2.27%.
- The general-equilibrium counterfactual has a welfare gain of 4.38%, nearly twice as large.
- Robustness rows show that the main welfare conclusion survives alternative modelling assumptions.

Economic message: The commodity super-cycle raises welfare, but labour-market transmission substantially reduces the size of the welfare gain.

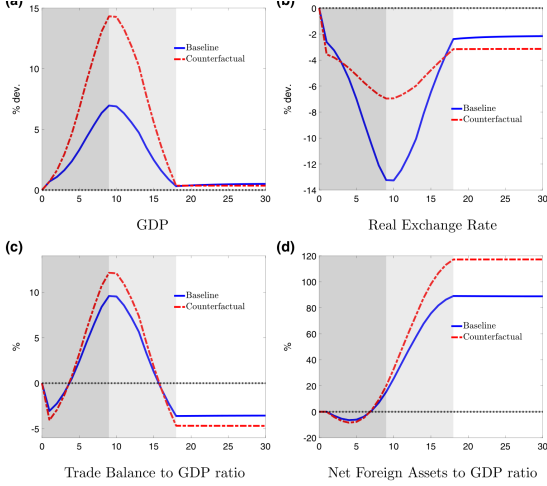


FIGURE 6
Response of macroeconomic outcomes to a commodity price super-cycle
Note: This figure shows the response of consumption-deflated GDP (top left), the real exchange rate (top right), the trade balance to GDP ratio (bottom left), and the net foreign assets to GDP ratio (bottom right) to the commodity price super-cycle shown in Figure 3. The blue solid line corresponds to the baseline economy and the red dashed line corresponds to the counterfactual economy described in

Figure 6: macroeconomic response

Welfare and long-run net foreign assets		
	CEQ Welfare (%)	Long Run B/Y
Baseline	2.27	0.89
GE Counterfactual	4.38	1.17
Multiregion model (Online Appendix B.1)	1.86	0.84
Rep. Firm model (Online Appendix B.2)	2.02	1.06
$B_0/Y = 0.20$ (Online Appendix B.3)	2.19	1.07
$B_0/Y = -0.20$ (Online Appendix B.3)	2.37	0.71
Bond-holding cost (Online Appendix B.4)	2.25	0
High f_x (Online Appendix B.5)	2.10	0.92
Low f_x (Online Appendix B.5)	2.49	0.85

Notes: This table reports consumption-equivalent welfare (Column 1) and the long-run value of the net foreign assets to GDP ratio (Column 2) to the commodity price super-cycle shown in Figure 3. The rows correspond to the baseline economy (first row), the counterfactual economy described in Section 4.3 (second row), the model with a representative firm in the tradable and nontradable sectors (third row), the multiregion model (fourth row), the baseline economy with an initial value of net foreign assets to GDP equal to 0.2 or -0.2 (fifth and sixth rows), the baseline economy adding a bond-holding cost (seventh row), and the baseline economy with a higher or lower value of the fixed cost of exporting

Table 8: welfare and long-run net foreign assets

8 Short summary

- The paper studies commodity price super-cycles using Brazilian regional and firm-level data.
- It identifies two transmission channels: a wealth channel and a cost channel.
- The wealth channel increases local demand and benefits firms selling locally.
- The cost channel raises wages, especially unskilled wages, and changes relative production costs.
- A regional commodity price index based on base-period commodity employment shares identifies local exposure.
- Regions with stronger commodity price increases experience larger declines in the skill premium.
- High-skill firms grow more than low-skill firms during regional commodity booms.
- Nonexporters and low-tradability manufacturing industries grow more than exporters and high-tradability industries.
- A dynamic heterogeneous-firm small open economy model quantifies the channels.
- Shutting down labour-market transmission greatly reduces reallocation and raises welfare.
- The main lesson is that commodity booms have both wealth benefits and distortionary labour-market effects.

9 Conclusion

9.1 Core conclusion

Benguria, Saffie, and Urzua show that commodity price super-cycles transmit through both demand and cost channels. The wealth channel raises domestic demand and helps local sellers. The cost channel raises wages, especially for unskilled labour, and changes relative production costs across industries.

9.2 Economic intuition

A commodity boom increases national and regional income, but it also pulls labour into the commodity sector. Because commodities are relatively unskilled-intensive, this increases demand for unskilled labour and lowers the skill premium. Local sellers benefit from higher spending, while exporters and low-skill-intensive industries face stronger cost pressure.

9.3 Takeaway

Commodity booms are not simply positive endowment shocks. Their aggregate welfare impact depends on how they reshape labour markets, firm selection, exporting, and sectoral allocation. The paper’s central contribution is to identify these channels in micro data and quantify their aggregate and welfare consequences in a dynamic model.